Week 7: International Banking and Eurodollar Markets

FINA3020 at CUHK Business School

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Lecture Overview: Banks' Intermediation of International Capital Flows [Core Concept]

- Correspondent banking: Nostro accounts, Vostro accounts, and payment chains
- Offshore deposits: Eurodollars, offshore CNH, and non-US banks' dollar funding
- Trade finance: letter-of-credit funding and export credit agencies
- Multinational firms' banking relationships: credit lines, cash pooling, netting, and counterparty limits
- Balance-sheet mismatch, bank runs, and the in-class bank-run game
- Non-bank intermediaries compared with banks
- Microprudential and macroprudential regulation, Basel III, and the financial trilemma

How Banks Intermediate Cross-Border Flows

[Core Concept]

International banks perform four core functions:

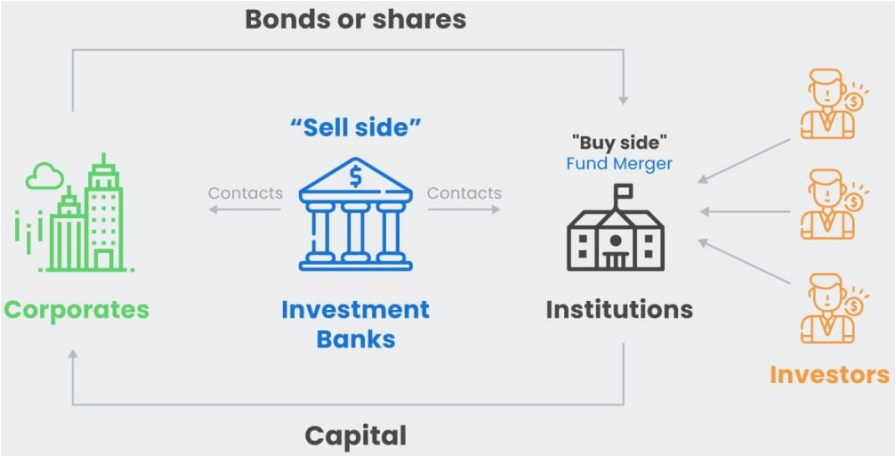
- **Maturity transformation**: convert short-term deposits into long-term loans across borders
- **Liquidity provision**: provide deposits payable on demand while holding illiquid loans
- **Currency market-making**: fund foreign-currency assets with domestic or offshore liabilities
- **Payment clearing**: settle trade and financial payments between countries

Investment Banking and Securities Markets [Core Concept]

- In primary markets, issuers sell new securities to raise cash. Investment banks advise on terms, underwrite or place the securities, and earn fees.
- In secondary markets, investors trade existing securities. Brokers match client orders, while dealers quote prices and hold inventory on their own balance sheets.
- Asset managers, pension funds, insurers, and private funds are buy-side investors. Universal banking groups operate both commercial and investment banks (banned in the US from 1933 to 1999, Glass-Steagall Act)
- Liquid secondary markets lower the cost of future issues. Secondary trades transfer cash between investors. The issuer receives cash only in the primary market.

Investment Banking and Securities Markets [Core Concept]

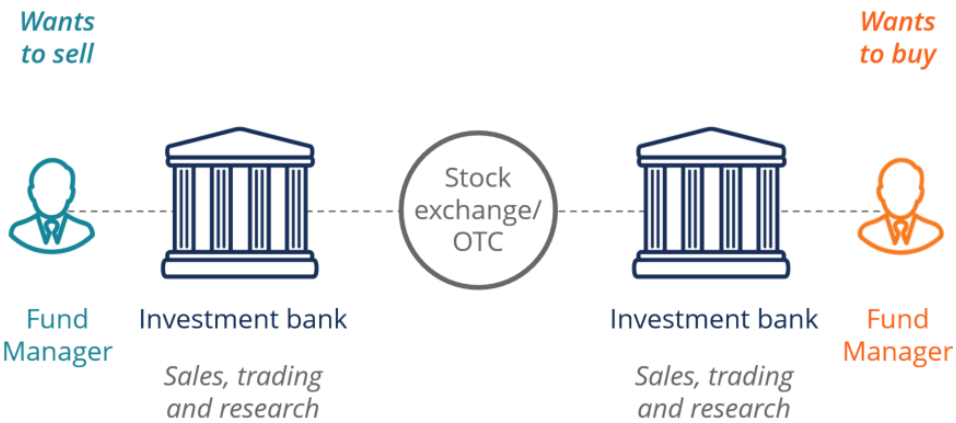
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Source: Srivastava.

Investment Banks Intermediate Capital Markets: Secondary Market

[Background Context]



Source: CFI.

Direct Markets vs. Bank Intermediation

Two main channels fund cross-border investment:

- **Direct capital markets**: issuers sell bonds or equities to investors, with underwriters arranging placement
- **Indirect bank intermediation**: banks take deposits and make loans, which they can hold, syndicate, or securitize

Capital Markets

- Exchanges or over-the-counter (OTC) dealers set prices
- End investors bear market and default risk
- Underwriting has high fixed costs

Banking Channels

- Banks screen and monitor borrowers directly
- Banks keep initial credit risk on their balance sheets
- Banks provide flexible credit lines and overdrafts

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Stylized Commercial Bank Balance Sheet

Assets	**$ Million**	**Liabilities + Equity**	**$ Million**
Cash and Central Bank Reserves	200	Retail Demand Deposits	1,600
Foreign Correspondent (Nostro)	300	Wholesale Foreign Deposits (Vostro)	1,200
Corporate Loans (Illiquid)	3,100	Short-Term Interbank Debt	800
Government Securities	400	Bank Capital (Equity)	400
Total Assets	**4,000**	**Total Liabilities + Equity**	**4,000**

- Capital ratio: $\frac{\text{Equity}}{\text{Assets}} = \frac{400}{4{,}000} = 10\%$
- Assets are long-term and illiquid
- Liabilities are short-term and payable on demand
- Equity absorbs losses when asset values fall

Example: The Bank Balance Sheet Balances [Core Concept]

All figures are in USD millions.

1. Add assets: cash/reserves 200 + correspondent balances 300 + loans 3,100 + government securities 400 = 4,000.
2. Add debt funding: retail deposits 1,600 + wholesale deposits 1,200 + interbank borrowing 800 = 3,600.
3. Equity is the residual: $4{,}000 - 3{,}600 = 400$. Assets equal liabilities plus equity.
4. A 100 loan loss, with liabilities unchanged, reduces assets to 3,900 and equity to 300. Equity absorbs losses first.

Interpretation

Equity absorbs asset losses, while cash and reserves pay withdrawals. A bank can be solvent (positive equity) and still run out of cash (illiquid).

Correspondent Banking Networks

[Core Concept]

Correspondent banking settles cross-border payments without a local branch:

- A respondent bank holds a deposit account at a correspondent bank in another country
- **Nostro account**: "our money at your bank" (recorded as an asset by the respondent bank)
- **Vostro account**: "your money at our bank" (recorded as a liability by the correspondent bank)
- Banks settle foreign-currency payments by debiting and crediting these interbank accounts

Example: Hong Kong Importer Pays a US Supplier

HSBC instructs JPMorgan to debit HSBC's dollar Nostro account (JPMorgan's Vostro) and credit the US supplier. No physical banknotes cross borders.

Types of correspondent banking → | Herstatt: events → | Settlement-loss calculation →

Example: Nostro and Vostro Accounting [Core Concept]

$1 million payment from a HK firm (HSBC customer) to a US supplier (JPMorgan customer):

HSBC Ledger (Respondent Bank)

- Liability: HK Firm Deposit Account (-$1,000,000)
- Asset: Nostro Account at JPMorgan (-$1,000,000)
- Nostro asset balance at JPMorgan decreases

JPMorgan Ledger (Correspondent Bank)

- Liability: HSBC Vostro Account (-$1,000,000)
- Liability: US Supplier Deposit Account (+$1,000,000)
- Vostro liability to HSBC decreases

Result: The payment settles by updating interbank account balances, without moving physical cash across borders.

Example: One Payment through Two Correspondents [Core Concept]

A Hong Kong garment importer pays a supplier in Bangladesh USD 100,000. The two local banks have no account with each other.



Bank	Fee (USD)	Amount left
Start		100,000
Correspondent A	15	99,985
Correspondent B	20	99,965
Supplier's bank	10	99,955

- **SHA**: each side pays its own bank. Banks in the chain deduct their fees. The supplier receives USD 99,955.
- **OUR**: the importer pays all fees. The supplier receives USD 100,000.
- Each bank checks sanctions and anti-money-laundering rules. Each check adds time. Week 10 compares this route with faster payment systems.

Correspondent banking diagram →

Forms of Foreign Bank Presence

Banks use four main legal structures to operate abroad:

- **Representative office**: handles client relations and market research; cannot take deposits or make loans
- **Foreign branch**: part of the parent bank; shares the parent's legal identity and capital
- **Foreign subsidiary**: separate local company with its own equity capital and board
- **Offshore banking unit**: unit that books nonresident or foreign-currency business under special host-country rules

Branch vs. Subsidiary Trade-off

A branch can move cash easily within the banking group. The host regulator can still keep the branch's local assets in the host country (ring-fencing). A subsidiary has separate legal liability and its own local capital. Parent guarantees and reputation can still spread its losses to the group.

Cross-Border Claims and Bank Exposure

[Background Context]

The BIS consolidated banking statistics classify claims by borrower residence, office location, and currency:

- **Cross-border claims**: loans or securities held by a bank office against a resident of another country
- **Local claims in foreign currency**: claims of a local branch or subsidiary on host-country residents in a foreign currency
- **Local claims in local currency**: claims of a local branch or subsidiary on host-country residents in the host currency

Global banking network structure:

- Global banks and nonbanks supply wholesale funding in dollars, euros, and other major currencies through loans, bonds, deposits, and FX swaps
- Borrowers with unhedged foreign-currency or short-term debt face higher rollover and solvency risk

What Is a Eurodollar Deposit? [Core Concept]

Eurodollar definition and core features:

- **Eurodollar**: a US dollar deposit booked at a bank office outside the United States (named after its European origin, not the euro currency)
- The same prefix applies to other currencies: Euroyen are yen deposits outside Japan, and Eurosterling are pound deposits outside the UK
- Banks accepting Eurodollar deposits are called **Eurobanks**

Key Characteristics:

1. The US FDIC does not insure offshore dollar deposits. Both home-country and host-country regulators supervise the booking bank
2. Historically, Eurodollar deposits avoided US reserve requirements. The Federal Reserve set reserve ratios on Eurocurrency liabilities to zero in 1990 and on all deposits in 2020
3. Contracts historically used LIBOR as the reference rate and now use SOFR (Secured Overnight Financing Rate) or other benchmark rates

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Origins of the Eurodollar Market

Historical catalysts behind offshore dollar growth:

- **Cold War asset protection (1950s)**: Soviet and Eastern bloc entities held dollar deposits at European banks in London and Paris to reduce the risk of US asset freezes
- **US Regulation Q**: capped the interest rates that banks in the US could pay on domestic deposits. Savers moved to London dollar accounts that paid more
- **US capital controls (1960s)**: the Interest Equalization Tax and foreign credit restraints made borrowing in New York expensive for foreign firms. Dollar lending moved offshore
- **Petrodollar recycling (1970s)**: oil-exporting countries deposited large dollar surpluses in London banks, which lent those dollars to importing countries

Regulation Q and Cold War History →

Eurodollar Deposit Creation Mechanics

[Core Concept]

Offshore dollar deposits grow through interbank transfers:

- A US depositor moves $10 million from a New York bank (Citi) to a London bank (Barclays)
- Barclays records a $10 million Eurodollar deposit liability to the depositor
- Barclays receives a $10 million Nostro deposit asset at its New York correspondent (Citi)
- Barclays lends $9 million to BMW and pays it into BMW's Citi account. Barclays now has a $9 million loan asset in place of $9 million of its Nostro asset
- Total offshore dollar deposits increase by $10 million. Onshore deposits at Citi stay at $10 million

2020 dollar shortage → Central-bank swap calculation →

Example: Eurodollar Deposit Creation T-Accounts [Core Concept]

Step-by-step T-account balances for offshore deposit expansion:

1. US Firm transfers $10M to Barclays (London)

   - **Barclays (London)**:
     - Asset: Nostro balance at Citi (NY) +$10M
     - Liability: Eurodollar Deposit +$10M
   - **Citi (New York)**:
     - Liability: US Depositor -$10M
     - Liability: Barclays Vostro +$10M

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Total deposit liabilities at Citi remain $10M (only the account holder changes from the US depositor to Barclays, and then $9M to BMW). Because both accounts are at Citi, Citi's reserve balance at the Federal Reserve stays unchanged.

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How Loans Create Deposits

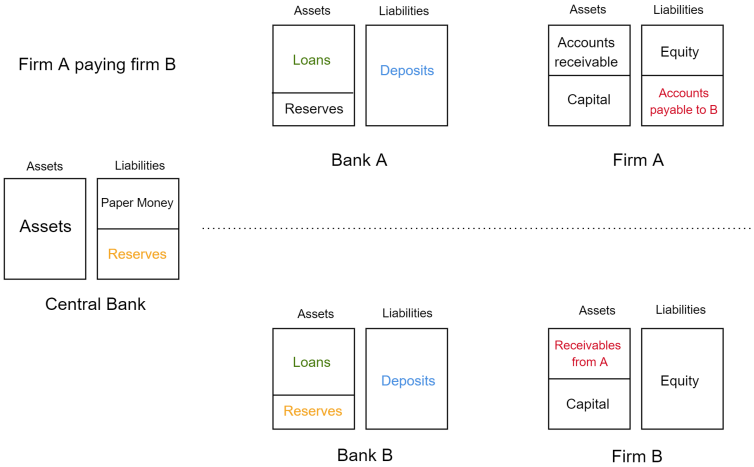
[Core Concept]

- When a bank makes a loan of 100, it credits the borrower's deposit account. Both loan assets and deposit liabilities rise by 100.
- When the borrower pays a customer at another bank, the lending bank transfers central-bank reserves or correspondent balances to the receiving bank.
- When the borrower repays principal, both the loan asset and the deposit liability fall. Interest payments reduce deposits and add to the bank's equity.
- Loan growth depends on borrower credit risk, bank capital, liquidity rules, and central-bank interest rates.

Source: Bank of England, Money creation in the modern economy (2014).

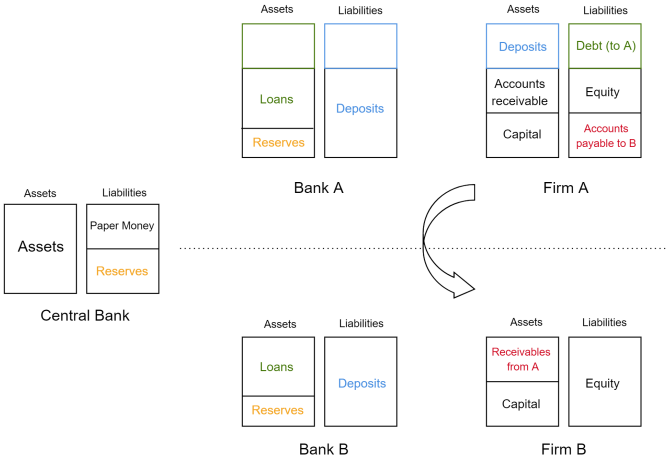
Money Creation by Commercial Banks

[Core Concept]



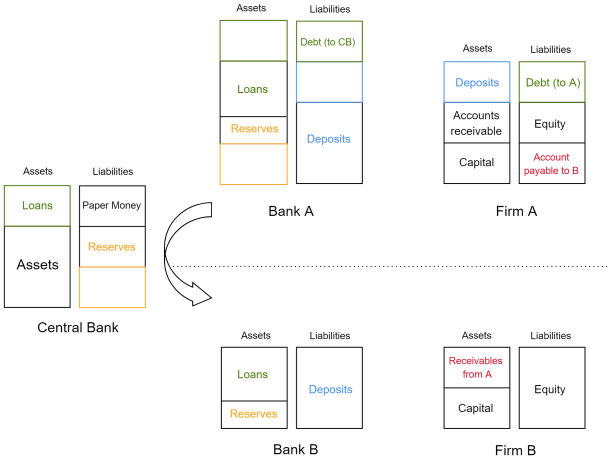
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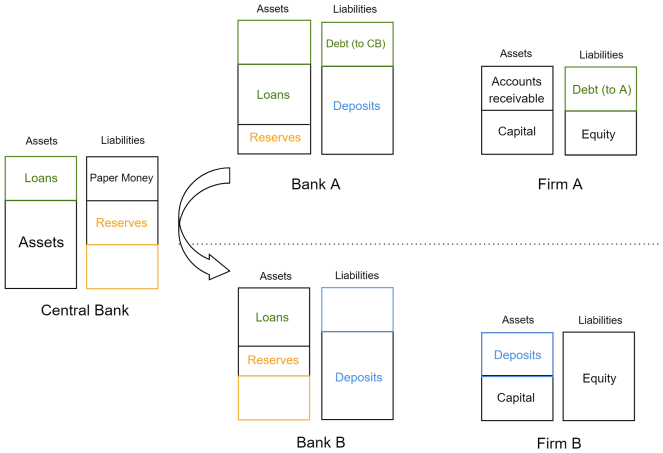
Money Creation by Commercial Banks

[Core Concept]



Money Creation by Commercial Banks

[Core Concept]



Growth of Offshore Dollar Intermediation

[Core Concept]

Development stages of offshore banking:

- **1960s**: US money-center banks opened London branches to fund international clients outside US interest-rate caps
- **1970s**: Hong Kong and Singapore grew into Asian Dollar hubs to finance regional trade and manufacturing
- **1980s–2000s**: Securitization and money market funds expanded wholesale cross-border dollar funding
- **Present day**: Foreign banks and borrowers hold trillions of dollars in offshore deposits, dollar loans, dollar bonds, and short-term FX swaps

Eurocurrency Classifications and Offshore Funding [Core Concept]

- A **Eurocurrency deposit** is a deposit booked outside the country that issues the currency. A USD deposit in London is a Eurodollar deposit for both a UK bank and a US bank branch.

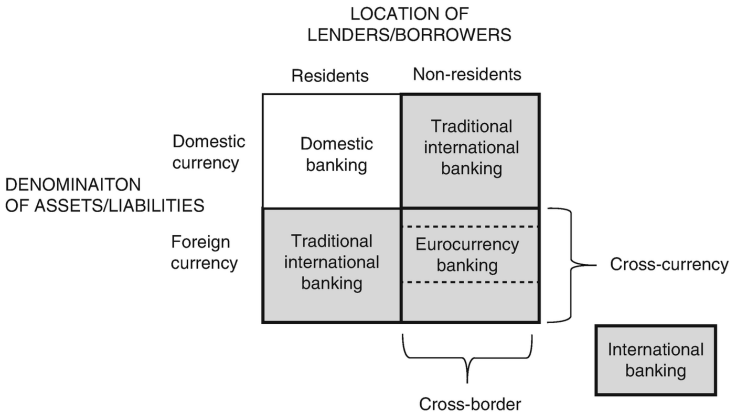
- **Onshore vs. offshore** compares the booking office location with the currency's home country. **Local vs. cross-border** compares the office location with the customer's residence.

- Multinational trade, dollar invoicing, and domestic banking rules drove Eurocurrency growth. Both home and host regulators supervise offshore offices.

- A domestic central bank can create its own currency, but it cannot print foreign currency. Without Federal Reserve swap lines or large dollar reserves, foreign banks face severe liquidity risk during a dollar run.

Eurocurrency Market Categorization

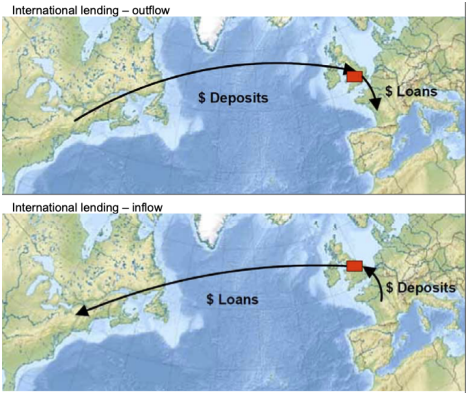
[Background Context]

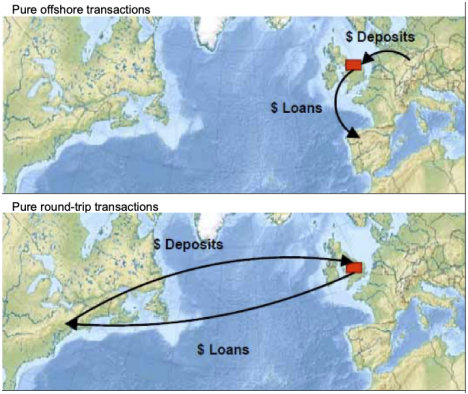


This diagram limits Eurocurrency banking to foreign-currency business with nonresidents. This course uses a broader definition: a deposit booked outside the currency's issuing country is offshore, whatever the depositor's residence. Customer residence (local or cross-border) is a separate distinction.

Source: Battilossi (2019).

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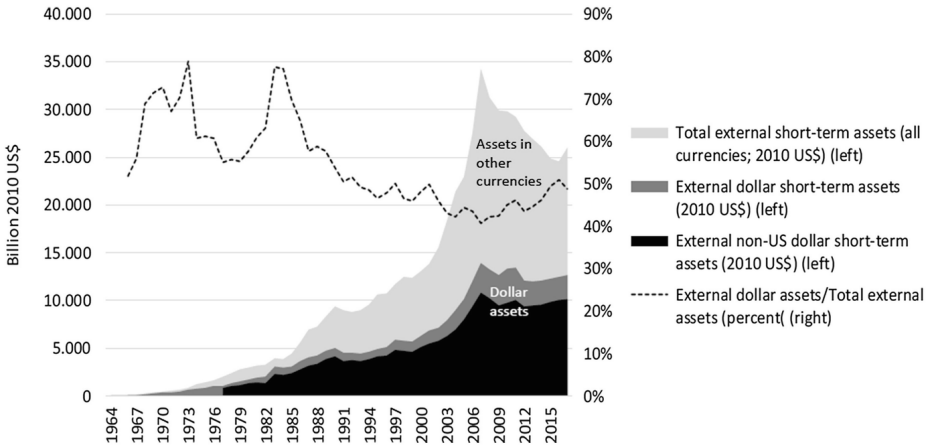




Source: He and McCauley (2012).

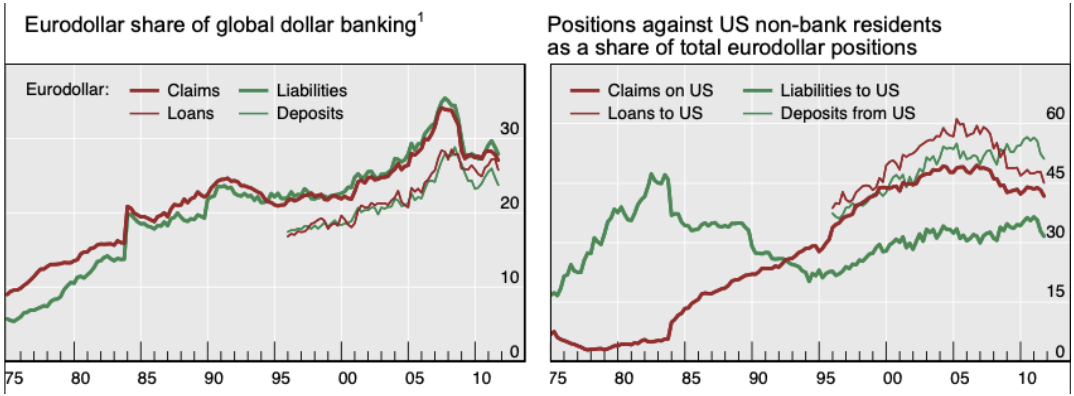
Eurocurrency Market Growth

[Background Context]



Source: Battilossi (2019).

Eurodollar Banking Growth



Source: He and McCauley (2012).

Exercise: Classify Five Deposits

[Core Concept]

Use two tests for each deposit:

- **Offshore**: the bank office is outside the country that issues the currency. Otherwise, the deposit is onshore.
- **Cross-border**: the depositor lives in a different country from the bank office. Otherwise, the deposit is local.

Deposit	Offshore?	Cross-border?
1. A Hong Kong trading firm holds USD at HSBC in Hong Kong	?	?
2. A Japanese insurer holds USD at Citibank in New York	?	?
3. A Hong Kong resident holds CNH at Bank of China (Hong Kong)	?	?
4. A US firm holds USD at the London branch of JPMorgan	?	?
5. A Singapore firm holds HKD at a bank in Hong Kong	?	?

Classify Five Deposits: Answers

[Core Concept]

Deposit	Onshore or offshore	Local or cross-border
1. Hong Kong firm, USD, HSBC Hong Kong	Offshore (Eurodollar)	Local
2. Japanese insurer, USD, Citibank New York	Onshore	Cross-border
3. Hong Kong resident, CNH, Bank of China (Hong Kong)	Offshore	Local
4. US firm, USD, JPMorgan London branch	Offshore (Eurodollar)	Cross-border
5. Singapore firm, HKD, bank in Hong Kong	Onshore	Cross-border

- The currency and the office location (not the depositor's identity) decide onshore or offshore.
- Deposit 4 shows that a US firm can hold Eurodollars. Deposit 2 shows that a foreign firm can hold onshore dollars.
- BIS locational statistics use the office location. They record deposits 2, 4, and 5 as cross-border liabilities of the bank.

Offshore CNH in Hong Kong

[Core Concept]

- CNH deposits at banks in Hong Kong are offshore deposits, like Eurodollars. At the end of July 2026, they totaled CNH 1,125.4 billion.
- **CNY** is the onshore currency. It trades in Mainland China under capital controls and a daily central parity. **CNH** is the same currency offshore. It trades freely in Hong Kong. Capital controls limit arbitrage, so the two rates can differ.
- Bank of China (Hong Kong) is the CNH clearing bank in Hong Kong. It connects offshore payments to the Mainland payment system (Week 10).

Example: An Offshore Liquidity Squeeze

On 12 January 2016, the overnight CNH HIBOR rose to 66.8%, from 13.4% one day before. Offshore CNH became scarce. A trader who borrowed CNH to sell the currency paid a very high rate for one day.

Sources: HKMA, Monetary Statistics for July 2026; GlobalCapital (2016).

A Non-US Bank's Dollar Balance Sheet

[Core Concept]

A Japanese bank's US dollar book, in USD billions:

Dollar assets		**Dollar liabilities**	
Loans to Asian firms	60	Customer deposits	35
US Treasury bills	20	Commercial paper sold to US money market funds (3 months)	35
Nostro balance at a US bank	20	FX swaps: yen swapped into dollars (1 month)	30
Total	**100**	**Total**	**100**

- The Bank of Japan can create yen. It cannot create dollars. The bank must get dollars from markets.
- **Shock 1**: money market funds do not roll over the 35 of commercial paper. The bank pays with the Nostro balance (20) and 15 of Treasury bills. It survives.
- **Shock 2**: swap dealers also stop the 30 of FX swaps. The bank must pay 65 but has only 40 of liquid dollars. It sells loans at a loss or borrows through a central-bank swap line.

March 2020 dollar shortage → Central-bank swap calculation →

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Overview Bank Functions Correspondent Banking Eurodollars Trade Finance Firms and Banks Bank Fragility Non-Banks Regulation

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Banking History and Hong Kong's Role

[Background Context]

- Early banks grew out of money changing, safekeeping, and trade credit. European merchant banks used bills of exchange, and Chinese remittance houses settled long-distance trade.
- The Bank of Amsterdam (1609) settled merchant payments in bank money. The Bank of England (1694) started as a private joint-stock bank that lent to the British government.
- After World War II, London became the center of Eurodollar wholesale funding. Hong Kong grew into a regional hub for trade finance, syndicated loans, and correspondent banking.
- Financial deregulation in the 1980s allowed commercial banks to expand into securities markets and securitization. This linked bank balance sheets to capital-market funding.

Early banking → | Settlement institutions → | Postwar banking → | Hong Kong tiers →

Letters of Credit: Which Bank Takes Which Risk? [Core Concept]

Week 5 showed the steps of a letter of credit (LC). This slide shows who bears risk:

Bank	What the bank does	Risk the bank takes
Issuing bank (importer's country)	Promises to pay against complying documents	The importer does not repay the bank
Advising bank	Checks that the LC is genuine and sends it to the exporter	No payment risk
Confirming bank (exporter's country)	Adds its own promise to pay	The issuing bank or its country does not pay
Discounting bank	Pays the exporter before the due date	No payment at maturity from the debtor bank

- An LC stays off the balance sheet until the bank pays. Basel rules count a short-term trade LC at 20% of its face value when they calculate capital (the credit conversion factor).
- Most trade finance is in US dollars. A non-US bank that pays a dollar LC needs dollar funding.

Capital for an LC →

Sources: Basel Framework, CRE20; BIS Bulletin No. 24 (2020).

Example: Discounting a Confirmed 180-Day LC

[Core Concept]

A **usance LC** lets the importer pay later. A bank can **discount** the LC: it pays the exporter now and collects from the issuing bank on the due date.

A Shenzhen exporter ships USD 1,000,000 of electronics to a buyer in Kenya. The buyer's Kenyan bank issues a 180-day usance LC. A Hong Kong bank confirms the LC and discounts it without recourse. Use ACT/360.

Item	Calculation	USD
Face value due on day 180		1,000,000
Discount at 6% per year	$1{,}000{,}000 \times 0.06 \times 180/360$	$-30{,}000$
Confirmation fee at 1% per year	$1{,}000{,}000 \times 0.01 \times 180/360$	$-5{,}000$
Cash to the exporter today		**965,000**

- Annualized cost on the cash received: $(35{,}000/965{,}000) \times (360/180) = 7.25\%$.
- The Hong Kong bank lends USD 965,000 for 180 days, funded with Eurodollar deposits.
- On day 180, the Kenyan bank pays USD 1,000,000. If the Kenyan bank does not pay, the Hong Kong bank takes the loss. The exporter keeps the cash.

Confirmed vs. unconfirmed →

Export Credit Agencies and Ex-Im Banks

[Core Concept]

An **export credit agency (ECA)** is a government agency or a government-owned bank. It supports exports from its home country.

- **Examples**: US EXIM; China Exim Bank and Sinosure; JBIC and NEXI (Japan); KEXIM and K-SURE (Korea); UK Export Finance; Hong Kong Export Credit Insurance Corporation.
- Private banks limit their exposure to risky countries and long maturities. The ECA takes the part of the risk that banks do not take.

Three main products:

1. **Cover for banks**: insurance or a guarantee if a foreign issuing bank does not pay an LC
2. **Working-capital guarantees**: cover for a bank's pre-shipment loan to the exporter
3. **Buyer credit**: a loan, or a guarantee of a bank loan, to a foreign buyer of capital goods

The OECD Arrangement on Officially Supported Export Credits sets minimum premiums, a minimum down payment (15% for most buyers), maximum repayment terms, and subsidy rules.

Some countries, for example China, do not participate.

How an ECA supports an LC → Buyer credit example →

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How an ECA supports an LC → Buyer credit example →

Example: An ECA Covers the Confirming Bank [Core Concept]

A bank confirms a USD 2,000,000 LC. A private bank in a high-risk country issues the LC.

- The confirming bank charges the exporter a 1.5% fee: USD 30,000.
- An ECA covers 95% of the loss if the issuing bank does not pay. The premium is 0.6%: USD 12,000.

Confirming bank's result (USD)	Without ECA cover	With ECA cover
Issuing bank pays	$+30{,}000$	$30{,}000 - 12{,}000 = +18{,}000$
Issuing bank defaults, no recovery	$30{,}000 - 2{,}000{,}000 = -1{,}970{,}000$	$30{,}000 - 12{,}000 - 100{,}000 = -82{,}000$

- In both cases, the confirming bank must pay the exporter USD 2,000,000. With cover, the ECA pays the bank $0.95 \times 2{,}000{,}000 =$ USD 1,900,000.
- Cover lets the bank confirm more LCs from risky countries. The ECA moves the default risk to home-country taxpayers.

Capital with and without cover →

Fee and premium rates are illustrative.

Trade Finance in Crises

[Core Concept]

- Banks fund LCs and trade loans with short-term money. Much of this money is in US dollars. When dollar funding becomes scarce, banks first cut credit lines to weak issuing banks.
- **2008–09**: at the London Summit (April 2009), G20 leaders promised USD 250 billion of trade finance support over two years. The support worked through export credit agencies and multilateral development banks.
- **2020**: trade finance depends heavily on dollars. A BIS study concludes that measures that ease dollar credit, such as swap lines, can protect global supply chains.
- A dollar funding shock to banks can become a trade shock for exporters in small economies.

Week 10: the dollar in invoicing and payments. Week 11: sudden stops, when foreign funding stops for a whole country.

March 2020 dollar shortage →

Sources: G20 London Summit Leaders' Statement (2009); BIS Bulletin No. 24 (2020).

What a Multinational Treasury Buys from Banks [Core Concept]

A multinational firm's treasury uses several banks. Each bank supplies some of these services:

- **Accounts and payments**: an operating account in each country, payroll, supplier payments, and collections
- **Liquidity**: a committed **revolving credit facility (RCF)**. Firm can draw and repay at any time
- **FX and hedging**: spot and forward FX, currency swaps, and interest-rate swaps (Week 4)
- **Trade finance**: letters of credit and guarantees (earlier in this lecture)
- **Capital markets**: underwriting the firm's bonds and equity, and M&A advice

The Relationship-Bank Group

A large firm has several **relationship banks**. These banks join the firm's syndicated RCF. In return, the firm gives them its fee-paying business: FX, cash management, and bond underwriting. Banks outside the RCF get little of this business.

Committed Credit Lines: Liquidity Insurance from Banks [Core Concept]

- A **syndicated RCF** is one contract with many lenders. Each bank commits a share. A 5-year term is common.
- The firm pays a **commitment fee** on the undrawn amount and **SOFR + margin** on the drawn amount. Rating agencies and commercial-paper investors expect a large firm to have one.

Example: A USD 1,000 Million RCF from 10 Banks

Commitment fee 0.20% per year on the undrawn amount. Drawn rate SOFR + 1.00%, with SOFR = 4%. Each bank commits USD 100 million.

- **Normal year, nothing drawn**: fee $= 0.002 \times 1{,}000 =$ USD 2.0 million, or USD 0.2 million per bank.
- **Stress year, USD 400 million drawn**: interest $0.05 \times 400 = 20.0$ plus fee $0.002 \times 600 = 1.2$. Total = USD 21.2 million.

The firm draws when other funding becomes expensive or closes. The banks must lend at the same moment.

Example: Why Banks Accept a Low Return on the Credit Line

Continue the RCF example. Look at one bank's USD 100 million share in a normal year.

1. **Capital**: assume the bank holds USD 5 million of equity against the commitment. Its cost of equity is 10% per year.
2. **Required profit**: $0.10 \times 5 =$ USD 0.5 million per year.
3. **Profit from the line**: commitment fee $0.002 \times 100 =$ USD 0.2 million. Return on equity $= 0.2/5 = 4\%$.
4. **Shortfall**: $0.5 - 0.2 =$ USD 0.3 million must come from other business with the firm.

The Relationship Trade

The firm buys cheap liquidity insurance. The bank expects FX, cash-management, and underwriting fees in return. A treasurer allocates this "wallet" to reward the banks that commit to the RCF, and tracks each bank's share every year.

Capital and cost of equity are illustrative.

Treasury Centres: Cash Pooling and Netting [Core Concept]

- **Treasury centre (in-house bank)**: one entity deals with the outside banks. Subsidiaries borrow from and deposit with the centre.
- **Cash pooling**: a bank sweeps each subsidiary's balance into one master account every day.
- **Multilateral netting**: subsidiaries report what they owe each other each month. Each one pays or receives only its **net** amount through the centre.
- **Benefits**: fewer FX trades and payments, less borrowing, and one view of group FX exposure (Week 4)
- **Limits**: China and India restrict cross-border pooling and netting. Intercompany loans face tax rules (Weeks 5 and 8).

Pooling Example

Singapore holds USD 50 million earning 3%. Mexico borrows USD 50 million from a bank at 7%. Pooling replaces the bank loan and saves about $0.04 \times 50 =$ USD 2 million per year.

Example: Multilateral Netting for Four Subsidiaries

Monthly payments between subsidiaries, in USD million equivalent. Each row is the payer.

Payer ↓ / Receiver →	Hong Kong	Germany	Singapore	US	Total paid	Total received	Net
Hong Kong	–	30	0	20	50	35	−15
Germany	10	–	15	0	25	35	+10
Singapore	0	5	–	20	25	25	0
US	25	0	10	–	35	40	+5
Total	35	35	25	40	135	135	0

- **Without netting**: 8 payments, USD 135 million gross.
- **With netting**: Hong Kong pays USD 15 million to the centre. The centre pays Germany USD 10 million and the US USD 5 million. Total USD 15 million.
- **Cost**: at 0.20% per payment for FX spread and bank fees, cost falls from $0.002 \times 135 =$ USD 0.27 million to $0.002 \times 15 =$ USD 0.03 million per month, saving USD 2.9 million per year.

Counterparty Limits: Where to Keep the Cash

[Core Concept]

- A deposit above the insurance limit (USD 250,000 in the US, HKD 800,000 in Hong Kong) is an unsecured claim on the bank.
- Treasury policy caps the cash at each bank and requires a minimum credit rating.
- Cash above the caps goes to Treasury bills or government money market funds.
- **SVB, March 2023**: Roku held about USD 487 million at SVB, about 26% of its USD 1.9 billion of cash. Most of it was uninsured.

Exercise: Apply a 15% Limit

Cash = USD 2,000 million.
Policy: at most 15% at one bank, and only banks rated A or above.
Limit = $0.15 \times 2{,}000 = 300$.

Bank	Rating	Balance	Excess
Bank A	AA	700	400
Bank B	A	400	100
Bank C	A	300	0
Bank D	BBB	200	200
Money fund	AAA	400	–

The treasurer moves USD 700 million to other A-rated banks, T-bills, or the money fund.

SVB case →

Firms and Banks in a Crisis: Credit-Line Drawdowns [Core Concept]

- **2008**: after Lehman failed, short-term creditors ran on banks. At the same time, firms drew down their credit lines. New loans to large US borrowers fell 47% in 2008Q4 from the previous quarter. Banks that shared more credit lines with Lehman faced more drawdowns and cut new lending more.

- **March 2020**: US firms drew on their lines at the fastest pace on record. The largest banks served most of the demand. Deposit inflows and Federal Reserve liquidity funded the loans.

- **2023**: firms with uninsured cash at SVB could not use it when the bank closed on 10 March. Regulators guaranteed all deposits on 12 March.

Sources: Ivashina and Scharfstein (2010), *Journal of Financial Economics*; Li, Strahan, and Zhang (2020), NBER WP 27256.

Example: A Credit-Line Drawdown on a Bank's Balance Sheet

Classroom bank, in USD millions. Liquid assets 40 (0% risk weight), loans 60 (100% risk weight), deposits 90, equity 10. Firms hold undrawn credit lines of 30 from this bank.

1. **Before**: capital ratio $= 10/60 = 16.7\%$. Liquid assets $= 40$, or $40/90 = 44\%$ of deposits.
2. **Firms draw 20 and move the cash to other banks**: liquid assets fall to $40 - 20 = 20$. Loans rise to $60 + 20 = 80$. Capital ratio $= 10/80 = 12.5\%$. Liquid assets $= 20/90 = 22\%$ of deposits.
3. **Depositors also withdraw 15**: liquid assets fall to $20 - 15 = 5$. Deposits fall to 75. Liquid assets $= 5/75 = 7\%$ of deposits.
4. **Result**: the bank has less cash for both groups and more risk-weighted assets. It cuts new lending to other borrowers.

In March 2020, many firms kept the drawn cash as deposits at the same large banks. That inflow offset much of the drain.

Source: Mechanism: Ivashina and Scharfstein (2010); Li, Strahan, and Zhang (2020).

Maturity Mismatch and Liquidity Risk

[Core Concept]

Banking balance sheets contain three main mismatches:

- **Maturity mismatch**: short-term demand liabilities fund long-term illiquid assets
- **Liquidity transformation**: banks promise instant cash to depositors while holding illiquid loans
- **Currency mismatch**: banks borrow short-term foreign currency (USD) to fund long-term domestic assets (KRW, HKD)

SVB: events → Bond price → Equity and withdrawals →

Example: A Currency Move Wipes Out the Equity Buffer [Core Concept]

Illustrative bank: KRW 143 billion of assets, US$100 million of debt, and no FX hedge. Hold the KRW value of assets fixed.

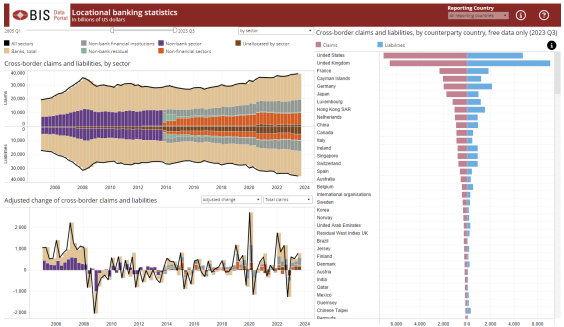
	Initially	After FX move
KRW per USD	1,300	1,430
USD debt in KRW billions	$0.1 \times 1{,}300 = 130$	$0.1 \times 1{,}430 = 143$
Equity in KRW billions	$143 - 130 = 13$	$143 - 143 = 0$

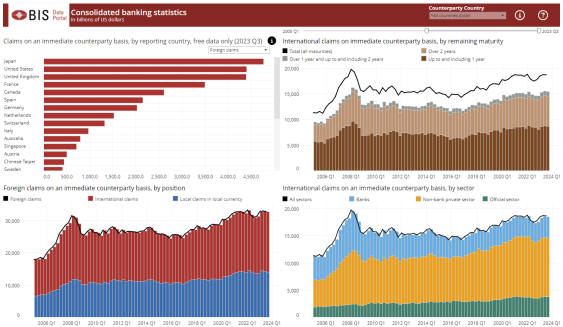
1. The USD/KRW rate rises 10%, so the dollar value of the won falls by $1 - 1{,}300/1{,}430 = 9.09\%$.
2. If the dollar debt is short-term while the KRW loans are long-term, the bank also faces rollover risk when the dollar debt matures.

- **Locational statistics** group claims by the country where the bank office is located. They follow balance-of-payments residency rules and track cross-border funding flows.
- **Consolidated statistics** group claims by the nationality of the parent bank and net out loans between offices of the same group. They measure a banking group's total country exposure.
- Example: a Japanese bank's London branch lends USD to a Brazilian firm. Locational data record a UK claim on Brazil; consolidated data record a Japanese bank claim on Brazil.
- Foreign banks lend either across borders from headquarters or locally through host-country affiliates. Cross-border loans usually contract faster than local affiliate loans during a crisis.

Patterns of Cross-Border Banking Activity

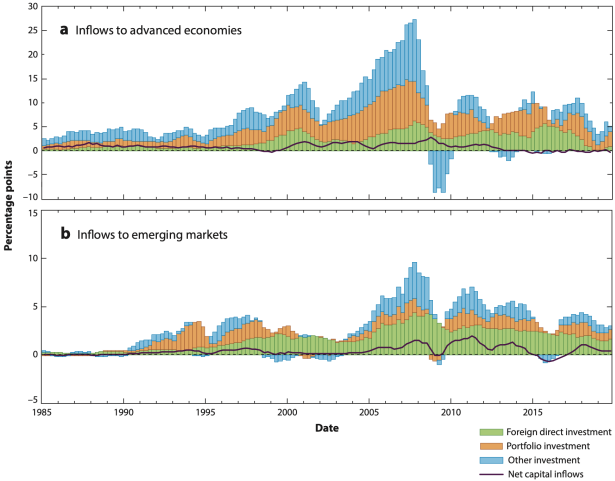
[Background Context]



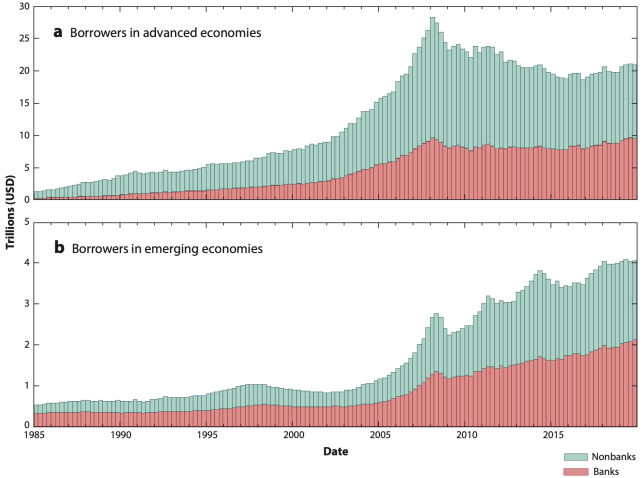


BIS locational data group positions by office location; consolidated data group positions by parent bank nationality.

Patterns of Cross-Border Banking Activity



Patterns of Cross-Border Banking Activity



Diamond and Dybvig (1983) Coordination Model

The baseline model of bank run fragility:

- Three periods: $t = 0, 1, 2$. Depositors face random liquidity needs at $t = 1$
- A long-term asset pays $R > 1$ at $t = 2$, while early liquidation at $t = 1$ yields only $L \leq 1$ (with $L = 1$ in the baseline model)
- The deposit contract provides liquidity insurance: early consumers receive c_1 at $t = 1$, and patient consumers receive c_2 at $t = 2$

Two Nash Equilibria:

1. **Good equilibrium**: impatient depositors withdraw at $t = 1$, and patient depositors wait until $t = 2$. The bank holds the remaining assets to maturity
2. **Run equilibrium**: both impatient and patient depositors withdraw at $t = 1$. The bank pays depositors in order of arrival (sequential service) and sells assets at a loss. It runs out of assets before it pays everyone

Numerical Payoff Matrix and Liquidation Math →

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Numerical Payoff Matrix and Liquidation Math →

A Bank Run: Cash Flows in Two Outcomes

Normalize deposits and initial assets to 1. The long asset returns 1.5 at maturity and 0.9 if liquidated early. Early consumers are 30%. Promised payouts are $c_1 = 1$ and $c_2 = 1.4$.

Outcome	Early liquidation	Available at maturity
Only early consumers withdraw	$0.3/0.9 = 1/3$	$(2/3)1.5 = 1.00$
All depositors run	All assets	0

- In the no-run equilibrium, the bank owes waiting depositors $0.7 \times 1.4 = 0.98$ at $t = 2$. The 1.00 maturity payoff covers this amount.
- In the run equilibrium, early liquidation yields only 0.90 at $t = 1$. The first 90% of depositors in line receive 1, and the last 10% receive 0.
- Waiting pays $1.4 > 1$ when other patient depositors wait, but pays $0 < 1$ when others run. Each patient depositor's best action depends on what others do.

Patient outcome → Run outcome →

Goldstein and Pauzner (2005) Global Games Model

Under the model's assumptions, including sufficiently small noise and dominance regions, noisy private signals select a unique equilibrium:

- Asset return depends on the bank's fundamental strength $\theta \in [0, 1]$.
- Each depositor i observes a noisy private signal $x_i = \theta + \varepsilon_i$ of bank strength.
- One-sided strategic complementarity: while the bank can still pay, a depositor's withdrawal incentive rises as more depositors withdraw. After the bank fails, more withdrawals lower that incentive.

Unique Cutoff Equilibrium:

- Under these assumptions, there is a unique signal cutoff x^*: depositor i withdraws if $x_i < x^*$ and waits if $x_i > x^*$.
- Weaker fundamentals θ lower the signals x_i, so more depositors receive signals below x^* and withdraw early.
- The bank fails when total withdrawals exceed its liquid resources. The probability of a run depends on bank fundamentals.

Threshold Cutoff Derivation →

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Threshold Cutoff Derivation →

Example: Private Signals and the Withdrawal Decision

Single-depositor payoff threshold: Assume a risk-neutral patient depositor receives 1 for sure by withdrawing at date 1. Waiting until date 2 pays 1.4 if the bank survives and 0 if the bank fails.

1. Let p be the depositor's belief that the bank survives after observing private signal x_i.
2. Expected payoff from waiting is $1.4p + (1-p)0 = 1.4p$.
3. The depositor prefers to wait when $1.4p \geq 1$, or $p \geq 1/1.4 = 71.43\%$.
4. At $p = 80\%$, waiting yields $1.12 > 1$ (wait). At $p = 60\%$, waiting yields $0.84 < 1$ (withdraw).

This simplified example illustrates how beliefs affect withdrawal decisions. The full model derives its cutoff from expected utility and other depositors' withdrawal choices.

Northern Rock and Korea: Two Funding Runs [Core Concept]

- **Northern Rock (UK, 2007)**: the mortgage bank relied heavily on securitization and wholesale funding. Wholesale lenders stopped rolling over funding in August 2007, before retail depositors lined up outside branches in September.
- Institutional lenders pulled funding first when mortgage-market liquidity fell. Retail depositors ran after the Bank of England announced emergency liquidity support.
- **Korea (1997)**: Korean banks borrowed short-term US dollars from foreign banks and lent to domestic firms. When foreign creditors doubted Korea's usable dollar reserves and corporate solvency, they stopped rolling over dollar loans.
- **Link to theory**: both runs combined weak fundamentals (funding mismatch or low usable reserves) with coordination pressure, where each creditor ran to avoid being last in line.

Northern Rock: events → | Funding calculation → | Korea: events → | Rollover shortfall → | Rollover recovery →

Comparing Bank Run Models

[Core Concept]

Diamond & Dybvig (1983)

- Multiple Nash equilibria (good vs. run)
- Self-fulfilling panic triggered by sunspots (shifts in expectations)
- A run can occur even when asset fundamentals remain unchanged
- Credible deposit insurance or lender-of-last-resort (LOLR) liquidity stops panics

Goldstein & Pauzner (2005)

- Unique equilibrium under small noise and dominance regions: depositors withdraw when their signal is below the cutoff x^*
- Run probability depends on asset fundamentals θ and signal noise
- Weaker fundamentals increase the share of depositors who withdraw
- The bank fails when fundamentals fall below a threshold θ^*. Higher capital and liquidity buffers lower θ^*, though guarantees can create moral hazard

Example: A Dollar Run on a Non-US Bank

[Core Concept]

Return to the Japanese bank's dollar book (USD billions): loans 60, Treasury bills 20, Nostro balance 20; customer deposits 35, commercial paper 35, FX swaps 30. The dollar book is matched. The bank's total equity of USD 8 billion is on its yen balance sheet.

September 2008: Lehman fails, and the Reserve Primary Fund's share value falls below USD 1. Investors pull cash out of prime money market funds. The 15 largest prime funds held half of their portfolios in claims on non-US banks. The funds stop buying bank commercial paper.

Next 30 days (USD billions)	Case A: roll half the swaps	Case B: roll no swaps
Outflow: commercial paper + FX swaps + deposits	$35 + 15 + 5 = 55$	$35 + 30 + 5 = 70$
Shortfall after using 40 of liquid dollars	15	30
Book value of loans sold at 75%	20	40
Fire-sale loss	5	10
Equity after the run	**3**	**−2**

Sources: Baba, McCauley, and Ramaswamy (2009), *BIS Quarterly Review*, p. 67 (15 largest prime funds, mid-2008); SEC (2009). Bank amounts are illustrative.

Example: Which Run Model Fits the Dollar Run?

[Core Concept]

- The loans are sound in Cases A and B. The whole loss comes from selling them early.
- **Diamond–Dybvig**: the same bank survives in Case A and fails in Case B. The only difference is what the dealers expect. Each dealer pulls out because it expects the others to pull out. The run is self-fulfilling.
- **Goldstein–Pauzner**: suppose news shows that the Asian loans will lose 10%, or 6. Equity falls to $8 - 6 = 2$ before any run. More creditors receive bad signals and withdraw. Even Case A's fire-sale loss of 5 now leaves equity at $2 - 5 = -3$.

Case B with sound loans is a panic: the bank is solvent and short of dollars. With a 10% loan loss, the run follows the fundamentals: the bank is short of dollars and of capital.

Example: Choose a Response to the Dollar Run [Core Concept]

Response	Result
Sell loans at 75% of book value	Loss of 5 (Case A) or 10 (Case B)
Swap yen into dollars	The price of dollars rises because every non-US bank wants them at once
Borrow dollars from the Bank of Japan, funded by a Fed swap line	No fire sale. The bank keeps its loans
Raise new equity	Needed when the loans lose value

The Bank of Japan can create yen. It cannot create dollars. A dollar lender of last resort stops a panic like Case B (roll no swaps). When the loans lose value, the bank also needs new capital.

March 2020 swap lines → Swap calculation →

Source: Baba, McCauley, and Ramaswamy (2009): in 2008, policy stopped the run on money market funds and replaced private dollar funding with public funding, partly through cooperating central banks.

In-Class Game: Bank Run Coordination

[Core Concept]

Work individually. Enter your CUHK ID, name, and section.

1. **Setup**: read the bank's assets and the deposit contract
2. **Diamond–Dybvig**: choose **Withdraw Early ($t = 1$)** or **Wait ($t = 2$)**, first when you expect others to wait, then when you expect them to run
3. **Goldstein–Pauzner**: observe a private signal $x_i = \theta + \varepsilon_i$ in three cases and choose whether to withdraw
4. **Runs and failures**: separate the pressure to pay depositors now from the quality of the loans
5. Submit your reflection



Open Week 7 Game

Non-Bank Intermediaries [Core Concept]

- **Non-bank intermediaries** also move savings to borrowers. Examples: money market funds, asset managers, pension funds, insurers, and hedge funds. Week 9 studies them.
- They do not take insured deposits. They have no direct access to central-bank loans.
- Many of them are linked to banks:
  - Money market funds lend to banks. They buy bank commercial paper (see the non-US bank's dollar balance sheet).
  - Hedge funds borrow from banks. A bank's **prime broker** unit lends to them through repo and margin loans against collateral.
- The lender gives less cash than the collateral value. The difference is the **haircut**. A higher haircut forces the borrower to add equity or sell assets.

Banks and Non-Bank Intermediaries Compared [Core Concept]

	Bank	Non-bank (for example, a hedge fund)
Short-term funding	Deposits, wholesale funding	Repo, prime-broker loans, investor money
Run trigger	Depositors withdraw	Lenders raise haircuts or call margin. Investors redeem
Forced sale	Loans and bonds	Securities, often the same positions that other funds hold
Backstop	Deposit insurance, lender of last resort	None directly. Central banks may buy the assets (March 2020)
Rules	Basel III capital and liquidity	Mainly through the lending banks: margin, haircuts, central clearing

Key Analytical Takeaway

Short-term debt that funds long-term or illiquid assets creates run risk in any intermediary. Bank loans to non-banks link non-bank stress to bank losses.

Microprudential vs. Macroprudential Regulation [Core Concept]

Two levels of financial stability regulation:

- **Microprudential supervision**: examines the safety and soundness of each individual bank in isolation
- **Macroprudential supervision**: targets system-wide risk, fire-sale spillovers, and credit cycles across the financial system

Microprudential Focus

- Capital ratios of each individual bank
- Bank-specific loan quality
- Safety of a single institution

Macroprudential Focus

- Countercyclical capital buffers
- Links across banks and nonbanks
- Preventing system-wide fire sales

Example: A Fire Sale

One bank can sell bonds for cash, and market prices stay the same. If all banks sell bonds at the same time, bond prices fall. Every bank that holds bonds then has mark-to-market losses. This is a fire sale.

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Microprudential vs. Macroprudential Regulation [Core Concept]

Two levels of financial stability regulation:

- **Microprudential supervision**: examines the safety and soundness of each individual bank in isolation
- **Macroprudential supervision**: targets system-wide risk, fire-sale spillovers, and credit cycles across the financial system

Microprudential Focus

- Capital ratios of each individual bank
- Bank-specific loan quality
- Safety of a single institution

Macroprudential Focus

- Countercyclical capital buffers
- Links across banks and nonbanks
- Preventing system-wide fire sales

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The Basel Accords Framework

[Core Concept]

Evolution of international bank capital standards:

- **Basel I (1988)**: set a minimum ratio of total capital to risk-weighted assets (RWA) of 8%
- **Basel II (2004)**: refined risk weights and let large banks use internal rating models, which understated risk before 2008
- **Basel III (2010–2017)**: raised minimum Common Equity Tier 1 (CET1) to 4.5% of RWA, plus capital buffers. It added a 3% leverage ratio (Tier 1 capital divided by total exposure, including off-balance-sheet exposures) and liquidity ratios

Basel III Liquidity Requirements:

- **Liquidity Coverage Ratio (LCR)**: high-quality liquid assets must cover 30 days of net cash outflows
- **Net Stable Funding Ratio (NSFR)**: available stable funding must cover required stable funding over 1 year

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Example: Risk-Weighted Assets and the CET1 Ratio

Use the stylized balance sheet (USD millions). The bank also has a USD 200 million trade LC outstanding. Assume all equity is Common Equity Tier 1 (CET1).

Item	Amount	Risk weight	Risk-weighted assets
Cash and central bank reserves	200	0%	0
Nostro balances (claims on banks)	300	20%	60
Corporate loans	3,100	100%	3,100
Government securities	400	0%	0
Trade LC (off balance sheet): $200 \times 20\%$	40	100%	40
Total			**3,200**

- CET1 ratio $= 400/3{,}200 = 12.5\%$. The minimum plus the conservation buffer is $4.5\% + 2.5\% = 7.0\%$, or USD 224 million. The excess is USD 176 million.
- The excess can support $176/0.07 =$ USD 2,514 million of new loans with a 100% risk weight.
- Leverage ratio $= 400/(4{,}000 + 40) = 9.9\%$, above the 3% minimum.

Risk weights are simplified classroom assumptions. Actual weights depend on ratings and national rules.

Basel III Liquidity Ratios (LCR and NSFR)

1. **Liquidity Coverage Ratio (LCR)**:

$$\text{LCR} = \frac{\text{High-Quality Liquid Assets (HQLA)}}{\text{Total Net Cash Outflows over 30 Days}} \geq 100\%$$

- Level 1 HQLA includes central-bank reserves and qualifying sovereign bonds (counted at 100%). Level 2 assets (such as qualifying corporate bonds) face haircuts and caps

- The 30-day stress scenario assigns higher run-off rates to short-term wholesale funding than to insured retail deposits

2. **Net Stable Funding Ratio (NSFR)**:

$$\text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\%$$

- ASF assigns high weights to equity capital, long-term debt, and stable retail deposits, and low weights to short-term wholesale debt

- RSF assigns high weights to long-term illiquid corporate loans and low weights to liquid sovereign bonds

Example: Calculate LCR and NSFR

Bank A, LCR: A 30-Day Stress Horizon, Amounts in Millions

Eligible HQLA after haircuts $= 120$. Stressed outflows $= 150$ and eligible inflows $= 50$. Inflows are capped at $0.75 \times 150 = 112.5$, so all 50 count.
Net cash outflows $= 150 - 50 = 100$, so LCR $= 120/100 = 120\%$.

Bank B, NSFR: 1-Year Structural Funding with Illustrative Weights

Funding: equity capital 20 at 100%, stable retail deposits 100 at 95%, funding from financial institutions with maturity under six months 80 at 0%.
Available stable funding (ASF) $= 20 + 95 + 0 = 115$.
Assets: cash 20 at 0%, qualifying unencumbered, performing corporate loans with at least one year remaining 180 at 85%.
Required stable funding (RSF) $= 0 + 153 = 153$, so NSFR $= 115/153 = 75.16\%$.

These are two separate examples. Bank A passes the 30-day LCR test ($120\% \geq 100\%$). Bank B fails the 1-year NSFR test ($75.16\% < 100\%$) because short-term wholesale debt funds long-term loans.

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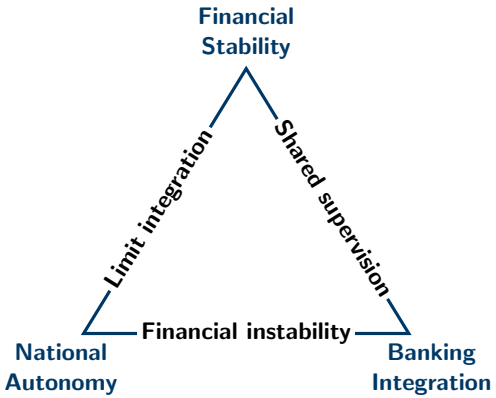
The Financial Trilemma (Schoenmaker, 2011)

[Core Concept]

Three desirable policy objectives:

1. **Financial stability**: preventing systemic bank collapses
2. **International banking integration**: open cross-border banking and capital flows
3. **National financial-policy autonomy**: national control over supervision, resolution, and bailout funds

Impossibility Result: A country can achieve at most two of these three goals at the same time. If its banks are integrated across borders, financial stability requires one of two actions: limit cross-border banking, or share supervision and fiscal backstops with other countries.



The Financial Trilemma in Practice

[Core Concept]

Keep these two	Illustrative example	What is given up
Stability + national autonomy	Bretton Woods-style national banking systems	Deep cross-border banking integration
Stability + banking integration	European Banking Union: ECB-led supervision and resolution through the Single Resolution Mechanism (deposit insurance remains national)	National supervisory discretion
Banking integration + national autonomy	Pre-2008 European cross-border banking with national supervisors	Financial stability in a system-wide crisis

When a cross-border bank gets into trouble, ask: **Who supervises it, who supplies emergency liquidity, and who absorbs the losses?**

Source: Schoenmaker (2011) and ECB Banking Supervision

Regulatory Incentives, Coordination, and Moral Hazard [Core Concept]

- **Regulatory arbitrage**: under risk-weighted capital rules, banks may buy the riskiest assets inside a low-risk bucket or shift exposures off the balance sheet. Unweighted leverage ratios and supervisory stress tests limit this behavior.
- **Cross-border resolution gaps**: global bodies (such as the Basel Committee) set common standards, while national governments control deposit insurance, bank resolution, and taxpayer backstops.
- **Moral hazard**: when creditors expect a government bailout, they charge banks lower interest rates and monitor risk less closely. Regulators counter this with extra capital surcharges for systemic banks and bail-in debt.
- **Resolving a failing bank**: authorities can protect depositors and keep payments running. Shareholders lose their equity, and unsecured bondholders take losses.

National reforms and moral hazard → Continental Illinois, 1984 →

Global Bank Transmission and Post-Crisis Retrenchment [Core Concept]

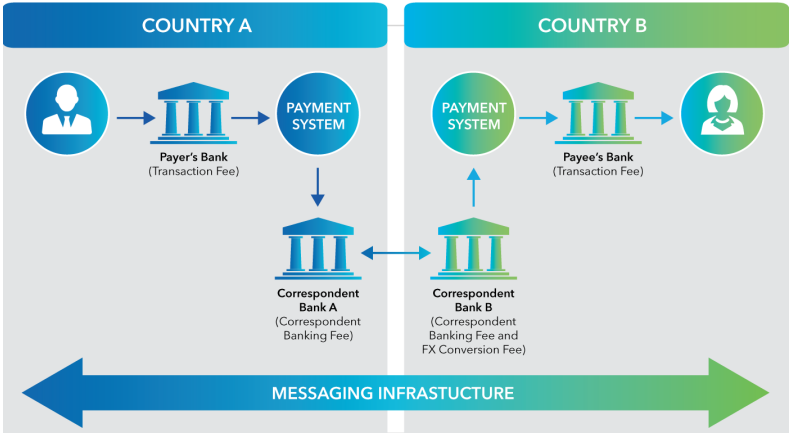
- When a parent bank suffers losses or funding stress at home, it often cuts lending at its foreign branches and subsidiaries. Margin calls and investor outflows transmit shocks across borders.
- After the 2008 crisis, European banks scaled back their global lending, while regional banks in Asia and other emerging markets expanded their cross-border share.
- Host regulators often require local subsidiaries to hold separate capital and liquidity buffers. This protects the host country but prevents the parent bank from moving spare cash across the group during a crisis.
- International banks also underwrite and trade offshore bonds (Eurobonds). This work connects bank credit, capital markets, FX hedging, and payment settlement.

Summary of International Banking [Core Concept]

- International banks perform maturity, liquidity, and currency transformation across borders
- Nostro and Vostro accounts clear cross-border payments through correspondent networks
- Eurodollar markets supply offshore dollar deposits and loans outside US FDIC insurance, supervised by home and host regulators
- Non-US banks fund dollar assets with short-term dollar debt. They cannot create dollars, so they face rollover risk
- Banks take the credit risk in letters of credit and can pay the exporter early by discounting the LC. Export credit agencies take risk that banks do not take
- Multinational firms buy credit lines from relationship banks, net and pool cash, and cap their exposure to each bank
- Liquidity transformation creates bank run fragility (Diamond-Dybvig sunspots vs. Goldstein-Pauzner fundamental thresholds)
- Non-bank intermediaries have no deposit insurance or central-bank loans. When their lenders raise haircuts, they must sell assets
- Basel III and macroprudential regulation manage systemic risk under the Financial Trilemma

Appendix: Correspondent Banking Diagram

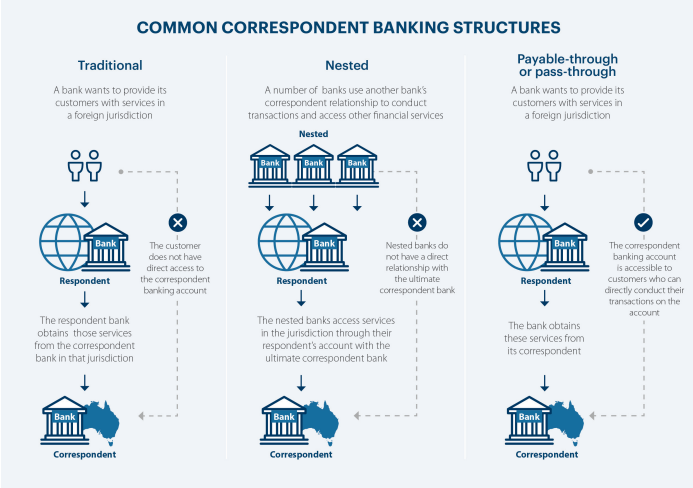
Source: Aite Group



← Back to Main Slide

Appendix: Types of Correspondent Banking

[Background Context]



← Back to Main Slide

Herstatt, 1974: An Unsettled FX Payment

Historical case: correspondent accounts settle the two currencies in different time zones.

- On 26 June 1974, German regulators closed Bankhaus Herstatt after counterparties had paid Deutsche marks in Frankfurt, but before Herstatt sent the corresponding US dollars in New York.
- Counterparties that had already sent Deutsche marks lost their paid principal when Herstatt closed. A trade at a locked-in FX rate still has settlement risk when the two currencies settle at different times.
- Other banks stopped outgoing payments until incoming funds arrived. International settlement and interbank credit froze.
- Payment-versus-payment (PvP) links the two legs so that one currency settles if and only if the other currency settles. CLS Bank began operating PvP settlement in 2002.

Discussion: Which risk comes from the exchange rate moving, and which comes from paying one currency before receiving the other?

Source: BIS, "FX settlement risk remains significant" (2019).

← Back to Main Slide Exposure calculation →

Herstatt Mechanism: Principal at Risk

[Background Context]

Classroom assumptions: Bank A buys US$10 million for DM25 million at 2.50 DM per USD. Bank A sends the marks first. Assume 40% eventual recovery on its dollar claim, a fixed exchange rate, and zero discounting or legal costs.

1. **Check the trade:** US$10m at 2.50 DM per USD = DM25m paid by Bank A.
2. **Counterparty fails before paying dollars:** immediate dollar receipt is 0. The unpaid principal is $10 - 0 =$ US$10m.
3. **Eventual recovery in bankruptcy:** $0.40 \times$ US$10m = US$4m.
4. **Credit loss:** US$10m − US$4m = US$6m. At 2.50 DM per USD, the loss equals $6 \times 2.50 =$ DM15m.

PvP solution: under payment-versus-payment (CLS), Bank A's DM25m is released only if Herstatt delivers the US$10m. Bank A cannot lose its principal.

Source: Mechanism: BIS (2019). Trade size, rate, and recovery are illustrative.

Case events

← Back to Main Slide

Appendix: Regulation Q and London Market Origins

[Background Context]

US Regulation Q Constraints:

- Enacted under the US Banking Act of 1933
- Capped the interest rates that banks in the US could pay on time and savings deposits
- Prohibited US commercial banks from paying interest on demand deposits

Offshore Regulatory Arbitrage:

- Banks operating in London were outside US Federal Reserve Regulation Q caps
- Paid higher interest rates on US dollar deposits than US law allowed domestic banks to pay
- Attracted global dollar deposits and built the London Eurodollar interbank market

← Back to Main Slide

Appendix: Diamond and Dybvig (1983) Model Setup

Mathematical Setup:

- Continuum of depositors of total mass 1, each depositing 1 unit at $t = 0$
- Fraction λ are impatient (early consumers, type 1); fraction $1 - \lambda$ are patient (late consumers, type 2)
- Investment technology: 1 unit yields $L \leq 1$ if liquidated early at $t = 1$ ($L = 1$ in the baseline model) or $R > 1$ at maturity $t = 2$

Optimal Risk Sharing vs. Bank Run:

- To insure depositors against liquidity risk, the deposit contract sets $1 < c_1^* < c_2^* < R$
- If patient (type 2) depositors expect everyone else to withdraw at $t = 1$:

  Total demand at $t = 1$ is $c_1^* > L \implies$ early liquidation cannot pay every depositor in full

- Waiting leaves nothing at $t = 2$, so running at $t = 1$ is each patient depositor's best response.

← Back to Main Slide

Appendix: Goldstein and Pauzner (2005) Threshold Derivation

Global Game Information Structure:

- Asset return $R(\theta)$ increases with the fundamental state θ
- Depositor i receives a private signal $x_i = \theta + \sigma\varepsilon_i$, where $\varepsilon_i \sim i.i.d.\ U[-1/2, 1/2]$
- As signal noise $\sigma \to 0$, all depositors' signals cluster tightly around θ

Equilibrium Cutoff Logic:

- Each patient depositor compares the payoff from withdrawing at $t = 1$ with the expected payoff from waiting until $t = 2$, given signal x_i and the share of other depositors expected to withdraw
- At the cutoff signal x^*, a patient depositor is indifferent between withdrawing and waiting:

$$\mathbb{E}[U_{\text{withdraw}} - U_{\text{wait}} \mid x_i = x^*;\ \text{others use } x^*] = 0$$

- Under the model's assumptions, including sufficiently small noise and dominance regions, this indifference condition gives a unique cutoff x^*. Stronger fundamentals θ reduce total withdrawals and lower the probability of a bank run.

← Back to Main Slide

Banking before Modern Banks

- **Ancient credit:** Mesopotamian clay tablets recorded grain and silver loans, and Hammurabi's Code regulated interest. Greek money changers and Roman bankers exchanged coins, held deposits, and made loans.
- **Chinese commercial finance:** merchant credit and remittance houses (*piu3 hou6, piao4 hao4* and *cin4 zong1, qian2 zhuang1*) took deposits, issued notes, and transferred funds across provinces long before modern joint-stock banks.
- **Medieval Italian commerce:** bills of exchange let merchants pay in one city and currency and settle in another. Merchants did not have to ship coins.
- **Clearing:** netting mutual obligations reduces the cash needed for settlement. Counterparties still have credit risk until final settlement.

Netting example

A owes B 100, B owes C 80, and C owes A 60. Net positions are A pays 40, B receives 20, C receives 20. Enforceable multilateral netting reduces gross payments of 240 to net payments of 40.

From Merchant Banks to Public Settlement Institutions

- Private banking families and partnerships financed trade and sovereign borrowers. Examples include Coutts in private banking and Barings and Rothschilds in merchant banking.
- Commercial banks such as Barclays and Lloyds grew through mergers, while joint-stock ownership expanded bank equity capital and national branch networks.
- Genoa's Casa di San Giorgio (1407) managed public debt and took deposits. Amsterdam's Wisselbank (1609) settled merchant payments through central ledger accounts.
- The Bank of England (1694) started as a privately owned bank that raised funds for the British government. It later took on lender-of-last-resort and note-issuing responsibilities and was nationalized in 1946.

Source: Bank of England, institutional history.

← Main slide Previous

Next →

Postwar Banking, Liberalization, and Hong Kong

- **Bretton Woods:** pegged exchange rates and capital controls limited private cross-border capital flows and preserved national monetary autonomy.
- **International expansion:** commercial banks followed multinational clients overseas through foreign branches, trade finance, and offshore Eurodollar funding. Hong Kong grew into a major regional banking center.
- **Deregulation:** the US phased out Regulation Q deposit-rate ceilings in the early 1980s, and London's 1986 "Big Bang" opened securities dealing to commercial banks.
- **Market-based credit:** securitization, repo, and money-market funds expanded wholesale funding beyond traditional deposits. This funding can run when lenders lose confidence or collateral loses value.

These changes increased cross-border liquidity. They also created new ways for wholesale funding to run.

← Main slide Previous Next →

Hong Kong Deposit-Taking Institutions: The Historical Tiers

Hong Kong's traditional three-tier system of authorized institutions:

Institution	Deposit-taking rules
Licensed bank	May accept deposits of any size and maturity, including current and savings accounts.
Restricted licence bank (RLB)	May accept deposits of HK$500,000 or more, with no maturity limit.
Deposit-taking company (DTC)	May accept deposits of HK$100,000 or more, with an original maturity of at least three months.

In August 2024, the HKMA announced a plan to simplify this into a two-tier system by merging deposit-taking companies into the restricted-licence tier over a five-year transition period.

Source: HKMA, consultation conclusions, 5 August 2024.

← Main slide Previous

Bank-Run Example, Step 1: The Patient Outcome [Core Concept]

Normalize total deposits and initial assets to 1. The long asset pays $R = 1.5$ at date 2 or $L = 0.9$ if liquidated early at date 1. Early consumers are 30% of depositors. Promised payments are $c_1 = 1$ and $c_2 = 1.4$.

1. Early withdrawal demand at date 1 is $0.30 \times 1 = 0.30$.
2. Each liquidated unit yields 0.9, so the bank liquidates $0.30/0.9 = 1/3$ of the asset.
3. The remaining $2/3$ of the asset matures to $(2/3) \times 1.5 = 1.00$ at date 2.
4. Patient depositors are owed $0.70 \times 1.4 = 0.98$ at date 2. The bank pays them in full and keeps 0.02.

Check: early consumers receive 1 at date 1, and patient depositors receive 1.4 at date 2. The bank stays liquid and solvent when only early consumers withdraw.

← Main slide

Next →

Bank-Run Example, Step 2: Everyone Demands Cash [Core Concept]

Keep the same assets and deposit contract, but suppose all depositors demand payment at date 1.

1. Total cash demanded at date 1 is $1 \times c_1 = 1$.
2. Liquidating the entire asset at date 1 yields only $1 \times L = 0.9$.
3. Under sequential service (first-come, first-served), the first 90% of depositors in line receive 1 each, and the last 10% receive 0.
4. If a patient depositor expects others to run, waiting yields 0 while running yields 1 with 90% probability. The fear of a run becomes self-fulfilling.

How policy stops the run

Suspending early withdrawals, insuring deposits, or providing central-bank lender-of-last-resort loans removes the incentive for patient depositors to run.

← Main slide Previous

Who Sets Rules, Who Supervises, and Who Bears Losses? [Core Concept]

- **United States:** the Dodd–Frank Act (2010) introduced annual stress tests, resolution plans (living wills), and the Volcker Rule limiting proprietary trading by banks.
- **Europe:** the European Union implemented Basel III through CRD IV/CRR and launched the Single Supervisory Mechanism (2014) under the ECB to supervise major euro-area banks.
- **Global standard setters:** the Basel Committee (BCBS) and Financial Stability Board (FSB) design global standards, which national legislatures and regulators adopt and enforce.
- **Bailouts vs. market discipline:** rescues in Continental Illinois (1984), Sweden (1992), and the 2008 crisis protected creditors and stopped runs. Rescues also encourage more risk-taking later, unless shareholders and subordinated creditors take losses.

Cross-border conflict: during a crisis, home and host regulators often compete to keep liquid assets inside their own borders and shift losses abroad.

← Main slide

March 2020: Dollar Funding across Borders [Core Concept]

Historical case: offshore dollar claims require dollar settlement liquidity.

- COVID-19 market stress disrupted global wholesale dollar funding in March 2020. Foreign banks outside the US needed dollars to roll over funding and meet payments.
- The Federal Reserve expanded central-bank dollar liquidity swap lines, adding temporary swap lines with nine more central banks (including the Bank of Korea) on 19 March 2020.
- In a swap line, the Fed lends US dollars to a foreign central bank and receives the foreign currency as collateral. The foreign central bank then lends those dollars to local banks.
- At maturity, the foreign central bank repays the dollars with interest at the original exchange rate and bears any default risk on its local banks.

Connection to Eurodollars: offshore banks can create dollar deposits on their books, but they need actual dollar balances in New York when customers withdraw or pay other banks.

Source: New York Fed, Logan (14 April 2020). Federal Reserve, Swap Lines FAQs.

← Back to Main Slide | Swap calculation →

Dollar Swap: Initial Exchange and Repayment

[Core Concept]

Classroom terms: a foreign central bank draws US$100m for 84 days at 1.25 USD per EUR, 1% annual simple dollar interest, ACT/360, and zero fees.

1. **Initial foreign-currency collateral:** US$100m $\div$ (1.25 USD per EUR) = EUR80m. The Fed provides US$100m and receives EUR80m.
2. **Year fraction:** $84/360 = 0.233333$ under the ACT/360 day-count convention.
3. **Dollar interest:** $100{,}000{,}000 \times 0.01 \times 84/360 =$ US$233,333.33.
4. **Settlement at maturity:** the foreign central bank pays the Fed $100{,}000{,}000 + 233{,}333.33 =$ US$100,233,333.33, and the Fed returns EUR80m at the locked-in 1.25 rate.

The foreign central bank owes the full dollar repayment to the Fed even if a local commercial bank defaults on its dollar loan.

Source: Structure and interest: Federal Reserve, Central bank liquidity swaps. Inputs are assumed.

Case events

← Back to Main Slide

Northern Rock, 2007: Wholesale Stress before the Queues [Core Concept]

Historical case: a mortgage bank funded by short-term wholesale markets.

- Northern Rock funded long-term UK mortgages mainly through securitized bonds, covered bonds, and short-term wholesale borrowing rather than retail deposits.
- In August 2007, global money markets froze and institutional lenders stopped buying Northern Rock's maturing debt. Long-term mortgages paid cash back only slowly.
- On 13 September 2007, news leaked that Northern Rock had asked the Bank of England for emergency liquidity support. Retail depositors queued outside branches the next day.
- On 17 September, the UK Treasury guaranteed existing deposits to stop the retail run, and nationalized the bank in February 2008.

Mechanism: wholesale funding stopped first. Then the news of emergency central-bank support started a retail deposit run.

Source: Treasury Committee (2008), funding model. timeline. Treasury Committee (2009), intervention.

← Back to Main Slide | Funding calculation →

Northern Rock: Measuring the Funding Shift

Reported figures: retail deposits were 62.7% of total liabilities and equity at end-1997, and fell to 22.4% (GBP 22.6B) at end-2006.

1. **Drop in retail funding share:** $22.4\% - 62.7\% = -40.3$ percentage points.
2. **Implied 2006 total liabilities and equity:** $22.6/0.224 \approx$ GBP100.89bn.
3. **Non-retail funding and equity:** $100.89 - 22.6 =$ GBP 78.29B, or $100\% - 22.4\% = 77.6\%$ of the balance sheet.
4. **Interpretation:** this GBP 78.29B combined securitized notes, covered bonds, wholesale money-market debt, and equity. The share maturing each week determined the immediate cash shortfall.

Discussion: Why does the maturity schedule of that GBP 78.29B determine how fast the bank runs out of cash?

Source: Treasury Committee (2008), paragraphs 16–17. Calculations use reported rounded figures.

Case events

← Back to Main Slide

Korea, 1997: Foreign Creditors Stop Rolling Debt

[Core Concept]

Historical case: short-term foreign-currency debt and usable reserve depletion.

- Korean banks borrowed short-term US dollars from foreign banks to fund domestic firms. When foreign banks refused to roll over maturing loans, Korean banks needed dollars immediately.
- The Bank of Korea had few usable dollar reserves left, and Korean firms defaulted. More creditors left. The Bank of Korea could print won, but won could not pay maturing US dollar debt.
- On 24 December 1997, the IMF and major governments agreed to pay official aid sooner. Foreign creditor banks agreed to roll over and extend short-term bank debt.
- External-debt rollover rates rose from 30–35% in December 1997 to over 95% by March 1998.

Mechanism: coordinating foreign banks to roll over loans stopped the immediate dollar outflow and gave Korean banks time to restructure.

Source: IMF IEO (2003), Korea and the rollover agreement. IMF Country Report 00/11 (2000), p. 36.

← Back to Main Slide | Shortfall → | Recovery →

Korea Mechanism: A Dollar Rollover Shortfall

Classroom bank: US$100m principal matures today. Usable dollar cash is US$20m. Assume a 35% rollover rate, no interest due today, and no other dollar inflows. Amounts are in US$ millions.

1. **Rolled-over debt:** $100 \times 0.35 = 35$. This remains a liability with a later maturity date.
2. **Principal payable in cash today:** $100 - 35 = 65$, or $100(1 - 0.35) = 65$.
3. **Dollar cash shortfall:** with only 20 of dollar cash available, the bank is short by $65 - 20 = 45$.
4. **Minimum rollover rate to avoid default today:** set cash payment $100(1 - r) \leq 20$, which gives $1 - r \leq 0.20$, or $r \geq 80\%$.

Decision: whenever the rollover rate falls below 80%, the bank must raise new dollars, sell dollar assets, or negotiate a standstill.

Source: Historical motivation: IMF Country Report 00/11, p. 36. Bank size and cash are assumed.

← Back to Main Slide Improved rollover →

Korea Mechanism: What Improved Rollover Changes

[Background Context]

Same classroom bank: US$100m matures today and usable dollar cash is US$20m. Raise the rollover rate to 95% (matching March 1998). Amounts are in US$ millions.

1. **Rolled-over debt:** $100 \times 0.95 = 95$. Cash principal payable today drops to $100 - 95 = 5$.
2. **Remaining dollar cash:** $20 - 5 = 15$. The bank pays non-rolling creditors in full with 15 left over.
3. **Compare with 35% rollover:** immediate cash outflow falls from 65 to 5, saving $65 - 5 = 60$ of dollar cash today.
4. **Remaining obligation:** rolled-over debt increases from 35 to 95 ($95 - 35 = 60$), which must still be repaid at future maturity dates.

Policy lesson: debt rollover solves the immediate liquidity shortage, while long-term solvency still requires sound loans and future dollar earnings.

Source: Historical motivation: IMF Country Report 00/11, p. 36. Cash flows are illustrative.

Previous calculation

← Back to Main Slide

SVB, March 2023: Rate Risk Meets a Fast Run

[Core Concept]

Historical case: unhedged interest-rate risk and concentrated uninsured deposits.

- Silicon Valley Bank invested rapid deposit inflows into long-term US Treasury and agency bonds. When the Fed raised interest rates in 2022, those bond prices fell sharply. Also, 94% of SVB's deposits were uninsured.
- On 8 March 2023, SVB announced that it had sold US$21bn of securities at a US$1.8bn after-tax loss. It also tried to raise new equity. This news worried its venture-capital depositors.
- Depositors withdrew over US$40bn on 9 March and requested over US$100bn more for 10 March. Regulators closed SVB on the morning of 10 March.
- On 12 March, US authorities used the systemic-risk exception to protect all depositors in full. Shareholders and some unsecured debtholders did not get this protection.

Mechanism: unrealized interest-rate losses became realized losses. Many uninsured depositors withdrew at the same time, and the bank had to sell bonds.

Source: Federal Reserve, SVB review (2023), table 1 and failure timeline. joint statement, 12 March 2023.

← Back to Main Slide | Bond calculation → | Equity and cash →

SVB Mechanism: A Bond Loses Value as Yields Rise

Classroom bond: a default-free 5-year zero-coupon bond pays US$100m at maturity. Suppose the market yield jumps from 2% to 5% (annual compounding).

1. **Pricing formula:** $P = F/(1+y)^T$, where $F = 100$ is face value, y is annual yield, and $T = 5$ years.
2. **Price at 2% yield:** $(1.02)^5 = 1.1040808$, so $P_0 = 100/1.1040808 = \text{US\$90.5731m}$.
3. **Price at 5% yield:** $(1.05)^5 = 1.2762816$, so $P_1 = 100/1.2762816 = \text{US\$78.3526m}$.
4. **Market-value loss:** $90.5731 - 78.3526 = \text{US\$12.2205m}$, a drop of $12.2205/90.5731 = 13.49\%$.

The bond still pays US$100m if held to maturity in year 5, but selling it today to meet withdrawals locks in the US$12.22m market-value loss.

Source: Case motivation: Federal Reserve, SVB review (2023). Bond and yields are assumed.

← Back to Main Slide | Balance-sheet effect →

SVB Mechanism: Equity Loss and a Cash Withdrawal

Same classroom bond, in US$ millions. Assume cash of 10, deposits of 90, and no hedging. Mark the balance sheet to market values.

1. **Before the rate rise (at 2% yield):** assets $= 10 + 90.5731 = 100.5731$. Equity $= 100.5731 - 90 = 10.5731$.

2. **After the rate rise (at 5% yield):** assets $= 10 + 78.3526 = 88.3526$. Equity $= 88.3526 - 90 = -1.6474$. The 12.2205 bond loss wipes out the 10.5731 of equity.

3. **Depositors withdraw 20:** the bank uses its 10 of cash and sells $20 - 10 = 10$ of bonds at market value (face value $10 \times (1.05)^5 = 12.7628$).

4. **After paying 20:** cash is 0, remaining bonds are worth $78.3526 - 10 = 68.3526$, deposits are $90 - 20 = 70$, and market-value equity is $68.3526 - 70 = -1.6474$.

The interest-rate shock reduced the market value of equity from $+10.57$m to -1.65m. Withdrawals then forced the bank to sell bonds and realize those losses in cash.

Source: Author's illustrative balance sheet. The calculation describes the mechanism, not SVB's actual portfolio or regulatory capital.

Bond calculation

← Back to Main Slide

Continental Illinois, 1984: Creditor Protection

[Core Concept]

Historical case: wholesale Eurodollar funding and the "too-big-to-fail" safety net.

- Continental Illinois funded rapid corporate loan growth with short-term wholesale borrowing, including Eurodollar deposits. Losses on energy loans weakened creditor confidence.
- In May 1984, institutional and foreign Eurodollar depositors ran electronically. Because hundreds of smaller US banks held correspondent deposits at Continental, regulators feared systemic contagion.
- On 17 May 1984, US regulators provided emergency liquidity and guaranteed all depositors and general creditors, including uninsured wholesale creditors.
- When outflows continued, the FDIC injected capital, took an 80% ownership stake, and replaced top management. Protecting uninsured creditors prevented contagion but created moral hazard.

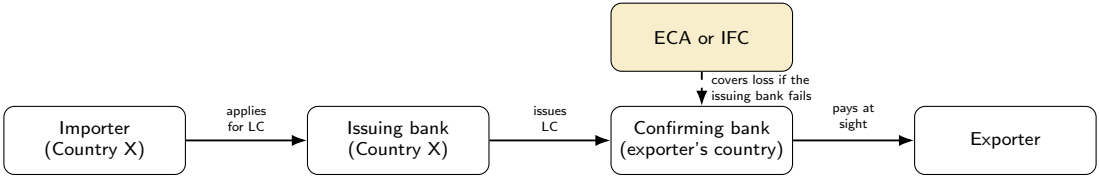
Discussion: How can a creditor guarantee stop a bank run today while weakening creditors' incentives to monitor bank risk in the future?

Source: Federal Reserve History, Continental Illinois (2023). FDIC, history of 1980–1989.

← Back to Main Slide

Appendix: How an Ex-Im Agency Supports an LC

[Core Concept]



- **US EXIM**: its LC policy covers a US bank that confirms a foreign bank's LC. Cover is 95% for a private issuing bank and 100% for a government-owned issuing bank.
- **IFC Global Trade Finance Program**: the World Bank's private-sector arm guarantees confirming banks against payment risk on emerging-market issuing banks. Cover can reach 100% of the transaction.
- With cover, the confirming bank accepts LCs from weaker issuing banks. Importers in risky countries can then buy on LC terms.

← Back to Main Slide Capital with and without cover →

Sources: US EXIM, Letter of Credit Insurance; IFC, Global Trade Finance Program.

Appendix: Buyer Credit for Capital Goods

[Core Concept]

A Vietnamese power company buys USD 100 million of turbines from a Korean manufacturer.

1. **Down payment**: the buyer pays 15%, or USD 15 million, in cash.
2. **Loan**: a commercial bank lends the buyer USD 85 million. The Korean ECA guarantees 100% of the loan for a 3% fee (USD 2.55 million).
3. **Payment to the exporter**: the bank pays the loan amount to the exporter at delivery. The exporter has no buyer credit risk.
4. **Repayment**: 10 equal semiannual principal payments of USD 8.5 million at a fixed 4% per year (2% per half-year).

First Payment and Total Interest

First payment: $8.5 + 85 \times 0.02 =$ USD 10.2 million. Total interest over five years: USD 9.35 million.

If the buyer defaults, the bank claims on the ECA. The ECA takes the credit and country risk.

← Back to Main Slide Full repayment schedule →

Appendix: Confirmed and Unconfirmed Discounting

Same LC: USD 1,000,000 due in 180 days, ACT/360. Without confirmation, a bank discounts the LC *with recourse* at 9% per year. It charges more because it relies only on the Kenyan bank.

USD	Confirmed, without recourse	Unconfirmed, with recourse
Discount	$1{,}000{,}000 \times 0.06 \times 0.5 = 30{,}000$	$1{,}000{,}000 \times 0.09 \times 0.5 = 45{,}000$
Confirmation fee	$1{,}000{,}000 \times 0.01 \times 0.5 = 5{,}000$	0
Cash to the exporter today	965,000	955,000
Annualized cost on cash received	7.25%	9.42%
Who loses if the Kenyan bank does not pay	Hong Kong confirming bank	Exporter (repays the bank)

- The confirmation fee is USD 5,000. It lowers the discount by USD 15,000 and removes the exporter's risk.
- The confirming bank knows the Kenyan bank through its correspondent relationship. It prices that risk better than a bank without the relationship.

All rates are illustrative.

← Back to Main Slide

Appendix: Capital for an LC with and without Cover

Use the USD 2,000,000 confirmed LC. Assume:

- Credit conversion factor for a short-term trade LC: 20%
- Risk weight on the foreign issuing bank: 100%. Total capital requirement: 8% of risk-weighted assets
- The ECA-covered part takes the risk weight of the ECA's government: 0%

USD	Without cover	With 95% ECA cover
Exposure at the issuing-bank risk weight	2,000,000	$0.05 \times 2{,}000{,}000 = 100{,}000$
Credit equivalent (20%)	400,000	20,000
Risk-weighted assets (100%)	400,000	20,000
Capital required (8%)	32,000	1,600

Cover cuts the capital for this LC by 95%. The 20% conversion factor already makes trade LCs cheap in capital compared with loans, which use 100%.

Assumptions are simplified. Source for the 20% factor: Basel Framework, CRE20.

← Back to Main Slide ← Back to Main Slide

Appendix: Buyer Credit Repayment Schedule

Loan USD 85 million. Principal USD 8.5 million every six months. Interest 2% per half-year on the opening balance. All figures in USD millions.

Payment	Opening balance	Interest (2%)	Principal	Total payment
1	85.0	1.70	8.5	10.20
2	76.5	1.53	8.5	10.03
3	68.0	1.36	8.5	9.86
4	59.5	1.19	8.5	9.69
5	51.0	1.02	8.5	9.52
6	42.5	0.85	8.5	9.35
7	34.0	0.68	8.5	9.18
8	25.5	0.51	8.5	9.01
9	17.0	0.34	8.5	8.84
10	8.5	0.17	8.5	8.67
Total		**9.35**	**85.0**	**94.35**

Interest falls by USD 0.17 million each period because the balance falls by USD 8.5 million. The buyer also pays the USD 2.55 million ECA fee.

← Back to Main Slide Previous